

Student 4 **Chintha Abhilash Reddy** 192525462RC

4 / 4 submitted

Q1. String Character Frequency Counter

CODE

```
s = input().lower().replace(" ", "")
freq = {}
for ch in s:
    freq[ch] = freq.get(ch, 0) + 1

for ch in sorted(freq):
    print(f"{ch} -> {freq[ch]}")
```

INPUT

Hello World

OUTPUT

```
d -> 1
e -> 1
h -> 1
l -> 3
o -> 2
r -> 1
w -> 1
```

Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()
vowels = set("aeiou")

v = {}
c = {}
for ch in s:
    if ch.isdigit() or ch.isspace():
        continue
    if ch.isalpha():
        if ch in vowels:
            v[ch] = v.get(ch, 0) + 1
        else:
            c[ch] = c.get(ch, 0) + 1

V = sum(v.values())
C = sum(c.values())

mf_v = max(v.items(), key=lambda x: x[1])[0] if v else "N/A"
mf_v_cnt = max(v.values()) if v else 0

lc_c = min(c.items(), key=lambda x: x[1])[0] if c else "N/A"
lc_c_cnt = min(c.values()) if c else 0

ratio = (V / C) if C else 0
print(f"Vowels: {V} | Consonants: {C} Most frequent vowel: {mf_v} ({mf_v_cnt}) "
      f"Least frequent consonant: {lc_c} ({lc_c_cnt}) "
      f"Vowel-to-Consonant Ratio: {ratio:.2f}")
```

INPUT

Python Programming

OUTPUT

Vowels: 4 | Consonants: 13 Most frequent vowel: o (2) Least frequent consonant: y (1) Vowel-to-Consonant Ratio: 0.31

7/29/26, 7:57 PM

Student Submissions — CO2_AT3_Coding Challenge

SIMATS Engineering • CSE • CSA0802 — Python Programming • 29-07-2026 • Category: CO2_AT3_Coding Challenge • Student Submissions Report

Q3. String Compression and Run-Length Encoding

CODE

```
def rle_encode(s):
    if not s:
        return ""
    out, i = [], 0
    n = len(s)
    while i < n:
        j = i
        while j < n and s[j] == s[i]:
            j += 1
        cnt = j - i
        out.append(s[i] + (str(cnt) if cnt > 1 else ""))
        i = j
    return "".join(out)

def rle_decode(t):
    if not t:
        return ""
    out, i = [], 0
    n = len(t)
    while i < n:
        ch = t[i]
        i += 1
        num = []
        while i < n and t[i].isdigit():
            num.append(t[i])
            i += 1
        k = int("".join(num)) if num else 1
        out.append(ch * k)
    return "".join(out)

s = input().strip()
enc = rle_encode(s)
compressed = enc if len(enc) < len(s) else s
print("Compressed:", compressed)
print("Decompressed:", rle_decode(compressed) if compressed != s else s)
```

INPUT aabbccdddee

OUTPUT Compressed: a2b3c2d4e2
Decompressed: aabbccdddee

7/29/26, 7:57 PM

Student Submissions — CO2_AT3_Coding Challenge

SIMATS Engineering • CSE • CSA0802 — Python Programming • 29-07-2026 • Category: CO2_AT3_Coding Challenge • Student Submissions Report

Q4. Advanced Text Statistics Analyser

CODE

```
import re
from collections import Counter

stop = {"the", "is", "a", "an", "in", "of", "and", "to", "for", "with", "on", "at", "by", "from", "as", "it", "this", "that", "or", "be", "are", "was", "we", "re"}

def syllables(word):
    w = re.sub(r'^a-zA-Z', '', word).lower()
    if not w: return 0
    groups = re.findall(r'[aeiouy]+', w)
    return len(groups) if groups else 1

text = input().strip()

paragraphs = [p for p in re.split(r'\n\s*\n', text) if p.strip()]
paras_count = len(paragraphs)

sentences = re.findall(r'[^\.\?]*[.!?]', text)
sent_count = len(sentences) if sentences else 1

words = re.findall(r'[A-Za-z']+', text.lower())
word_count = len(words)

avg_word_len = (sum(len(w) for w in words)/word_count) if word_count else 0
avg_sent_len = (word_count/sent_count) if sent_count else 0

freq = Counter(w for w in words if w not in stop)
top5 = sorted(freq.items(), key=lambda x: (-x[1], x[0]))[:5]

tot_syll = sum(syllables(w) for w in words)
syll_per_word = (tot_syll/word_count) if word_count else 0

flesch = 206.835 - 1.015*(word_count/sent_count) - 84.6*syll_per_word if word_count else 0

print(f"Words: {word_count} | Sentences: {sent_count} | Paragraphs: {paras_count}")
print(f"Avg Word Length: {avg_word_len:.2f} | Avg Sentence Length: {avg_sent_len:.2f}")
print("Top 5 Words: " + " ", ".join([f"{w}({c})" for w,c in top5]) if top5 else "Top 5 Words: ")
print(f"Flesch Reading Ease: {flesch:.1f}")
```

INPUT

Python is a powerful programming language

OUTPUT

Words: 6 | Sentences: 1 | Paragraphs: 1
Avg Word Length: 6.00 | Avg Sentence Length: 6.00
Top 5 Words: language(1), powerful(1), programming(1), python(1)
Flesch Reading Ease: 17.4